ETHEREUM

DEEP-DIVE RESEARCH SERIES

FILE 3 OF 3

ANALYSIS & OUTLOOK

Challenges & Controversies · Layer 2 Ecosystem
Competitive Landscape · Future Roadmap · Conclusion · References

Prepared by: NASIR BELLO

June 2025 · Personal Research Report · For Informational Purposes Only

TABLE OF CONTENTS

07 — CHALLENGES & CONTROVERSIES

Ethereum is the most battle-tested and widely used programmable blockchain in existence — but it is far from perfect. Over its decade-long history, it has faced fundamental technical limitations, governance crises, economic controversies, and regulatory challenges. Understanding these challenges honestly is essential to evaluating Ethereum's long-term prospects.

7.1 — The Scalability Problem & High Gas Fees

Ethereum's most persistent and consequential limitation has been its inability to scale at the base layer. Since the DeFi summer of 2020, Ethereum's base layer has consistently run at or near capacity — resulting in gas fees that have priced out ordinary users and made many use cases economically unviable.

The Gas Fee Crisis in Numbers

PERIOD	AVG GAS PRICE (GWEI)	SIMPLE ETH TRANSFER COST	UNISWAP SWAP COST
2019 (low usage)	5-10 Gwei	~$0.05	~$0.20
DeFi Summer 2020	100-200 Gwei	~$5-10	~$20-40
NFT Boom Jan 2022	200-500 Gwei	~$30-80	~$100-250
ETH ATH Nov 2021	150-300 Gwei	~$50-100	~$150-300
Bear Market 2023	10-30 Gwei	~$1-3	~$5-15
Post-Dencun 2024	3-15 Gwei	~$0.50-2	~$2-10 (L2: <$0.01)

At peak congestion, ordinary users were paying $100-$300 for a single Uniswap trade — making Ethereum DeFi accessible only to wealthy participants. This drove billions of dollars to alternative chains like Solana, BNB Chain, and Avalanche. The gas fee problem has been Ethereum's single greatest competitive weakness and the primary driver behind the entire Layer 2 scaling ecosystem.

Root Cause: The Block Size Debate

Unlike Bitcoin, where block size was a political battle, Ethereum's scaling limitation is technical. Ethereum's ~30 million gas limit per block is deliberately constrained to ensure that all nodes — including those running on modest hardware — can process blocks within the 12-second slot time. Increasing the gas limit would speed up the network but potentially centralise validation to only powerful servers. This fundamental tension between scalability and decentralisation is why Ethereum chose the Layer 2 scaling path rather than simply increasing base layer throughput.

7.2 — The DAO Hack & the Ethereum Classic Split

In June 2016, Ethereum faced its most existential crisis — one that fundamentally tested its governance model and left a permanent philosophical scar on the community.

What Was The DAO?

The DAO (Decentralised Autonomous Organisation) was a groundbreaking smart contract deployed on Ethereum in April 2016. It functioned as a decentralised venture fund — token holders could vote on investment proposals, and the DAO would automatically deploy funds according to the votes. It raised 12.7 million ETH — worth approximately $150 million at the time — making it the largest crowdfunding event in history at that point.

The Hack

On June 17, 2016, an unknown attacker exploited a reentrancy vulnerability in The DAO's smart contract — a bug that allowed them to repeatedly withdraw ETH before the contract updated its balance. The attacker drained approximately **3.6 million ETH (~$60 million)** into a child DAO. The funds were not immediately accessible due to a time-lock, giving the Ethereum community a narrow window to respond.

The Hard Fork Controversy

The Ethereum community faced an agonising choice: respect the principle that 'code is law' (the smart contract executed exactly as written, even if the outcome was theft) or intervene by hard forking the blockchain to return the stolen funds. After heated debate, the majority voted to hard fork — modifying Ethereum's history to move the stolen ETH to a recovery address.

- **The fork side (ETH):** Argued that protecting investors and the ecosystem's credibility outweighed the philosophical principle. The future of Ethereum was more important than ideological purity.
- **The anti-fork side (ETC):** Argued that blockchain immutability is the foundational promise — if the chain can be altered once for a political reason, it can be altered again. Ethereum Classic (ETC) was born as the unforked chain.
- **Long-term impact:** The fork succeeded in recovering the funds. Ethereum (ETH) went on to become the world's most valuable smart contract platform. Ethereum Classic persists as a minority chain. The debate about immutability vs. pragmatism remains unresolved in the broader crypto community.

The DAO's Legacy

The DAO hack permanently shaped Ethereum's smart contract development culture — introducing the concept of formal smart contract auditing as a necessity, not an option.

It demonstrated that smart contract bugs are not theoretical — they have real financial consequences at massive scale.

The reentrancy vulnerability exploited in The DAO is now one of the first things taught in every Solidity security course. The 'checks-effects-interactions' pattern was developed directly in response.

7.3 — Lido & Staking Centralisation

Since The Merge, Ethereum's most pressing decentralisation concern has been the dominance of **Lido Finance** in the liquid staking market. Lido controls approximately **28-32% of all staked ETH** — a concentration that raises serious questions about Ethereum's consensus security.

STAKING ENTITY	APPROX. ETH STAKED	% OF ALL STAKED ETH
Lido Finance	~9.5M ETH	~30%
Coinbase (cbETH)	~1.8M ETH	~6%
Binance (BETH)	~1.2M ETH	~4%
Rocket Pool	~800K ETH	~2.5%
Kraken	~700K ETH	~2.2%
Solo Stakers	~5M ETH	~16%
Other/Unknown	~13M ETH	~40%

The concern is straightforward: if Lido's node operators were to collude or be coerced (e.g., by government regulators), they could represent a meaningful fraction of the 33% threshold needed to disrupt Ethereum's finality. The Ethereum community has debated multiple responses including protocol-level staking caps, but none have been implemented due to concerns about restricting market activity.

7.4 — MEV: Maximal Extractable Value

Maximal Extractable Value (MEV) refers to the profit that block producers (validators) can earn by reordering, inserting, or censoring transactions within a block. On Ethereum, MEV has become a multi-billion dollar industry that has profound implications for user fairness and network decentralisation.

- **Front-running:** MEV bots monitor the mempool for large pending trades. When they see a large swap that will move a token's price, they insert their own transaction ahead of it to profit from the price movement they know is coming.
- **Sandwich attacks:** An attacker places a buy order before a user's large trade and a sell order immediately after — profiting from the price impact caused by the user's own trade.
- **Liquidation MEV:** When a DeFi loan becomes undercollateralised, bots race to be the first to liquidate it and claim the liquidation bonus.
- **MEV-Boost:** Flashbots' MEV-Boost infrastructure separates block building from block proposing — validators outsource block construction to specialised builders who maximise MEV extraction and share profits with validators. Over 90% of Ethereum blocks are now built via MEV-Boost.
- **PBS (Proposer-Builder Separation):** Ethereum's roadmap includes enshrining PBS at the protocol level — formalising the separation between validators (who propose blocks) and builders (who construct them) to reduce centralisation pressure.

7.5 — Regulatory Uncertainty

Ethereum faces unique regulatory challenges compared to Bitcoin. While Bitcoin is widely accepted as a commodity (similar to gold), ETH's regulatory status has been more contested:

- **The Howey Test:** The SEC has historically argued that ETH (particularly post-Merge) might constitute a security under the Howey Test, given that stakers earn returns from the efforts of others (the Ethereum Foundation and developer teams).
- **CFTC vs SEC:** The CFTC has classified ETH as a commodity, while the SEC's position has been ambiguous. The approval of spot ETH ETFs in 2024 by the SEC was widely interpreted as implicit acknowledgement that ETH is a commodity, not a security.
- **Staking regulations:** The SEC's actions against Kraken and Coinbase's staking products raised concerns about the legality of offering staking-as-a-service to retail investors.
- **Tornado Cash sanctions:** The US Treasury's sanctioning of Tornado Cash (an Ethereum privacy protocol) in 2022 raised serious questions about on-chain censorship and whether validators could be compelled to censor transactions from sanctioned addresses.

7.6 — Complexity & Developer Experience

Ethereum's post-Merge architecture — separating execution and consensus layers across multiple clients — has made the system significantly more complex to understand, develop for, and operate. Running a validator now requires running two separate client pieces of software that must communicate with each other. The proliferation of Layer 2s has fragmented liquidity and user experience. New developers face a steep learning curve navigating mainnet, testnets, multiple L2s, bridges, and the nuances of Solidity security.

08 — THE LAYER 2 ECOSYSTEM

8.1 — What Are Layer 2s and Why Do They Exist?

Layer 2 (L2) networks are blockchains that sit on top of Ethereum, inheriting its security while processing transactions off the main chain — then periodically settling the results back to Ethereum in batches. They are Ethereum's answer to the scalability problem: rather than making the base layer faster (which risks decentralisation), Layer 2s handle the throughput while Ethereum maintains decentralised security underneath.

The Two Main L2 Technologies

- **Optimistic Rollups:** Bundle thousands of transactions off-chain and post compressed transaction data to Ethereum. They assume all transactions are valid by default ('optimistically') and use a fraud-proof system — a 7-day challenge period during which anyone can submit a fraud proof if they detect an invalid transaction. Fast to compute but slow to withdraw (7 days for full security).
- **ZK Rollups (Zero-Knowledge Rollups):** Bundle transactions off-chain but generate a cryptographic proof (a ZK-SNARK or ZK-STARK) that mathematically proves all transactions are valid. This proof is verified on Ethereum, providing immediate cryptographic finality without a challenge period. More secure and faster withdrawals but computationally intensive to generate proofs.

EIP-4844: The Game Changer for L2s

In March 2024, Ethereum's Dencun upgrade introduced EIP-4844 — also called Proto-Danksharding.

EIP-4844 created a new transaction type called 'blobs' — temporary data chunks that L2s use to post their transaction data to Ethereum much more cheaply than before.

The result: L2 transaction fees dropped by 90-99% overnight. Transactions on Arbitrum, Optimism, and Base fell from cents to fractions of a cent.

This was the single most impactful upgrade for everyday Ethereum users since The Merge.

8.2 — Optimistic Rollups: Arbitrum & Optimism

Arbitrum One (ARB)

Optimistic Rollup

Arbitrum is the largest Ethereum Layer 2 by Total Value Locked — consistently holding $10-15B+ in TVL and processing more daily transactions than Ethereum mainnet itself. Developed by Offchain Labs, Arbitrum uses a unique multi-round fraud proof system (interactive fraud proofs) that is more efficient than single-round fraud proofs used by competitors. Arbitrum's EVM equivalence means virtually any Ethereum smart contract can deploy on Arbitrum with zero code changes. It has attracted the vast majority of Ethereum's DeFi ecosystem — Uniswap, Aave, Compound, GMX, Camelot, and hundreds of other protocols all have significant Arbitrum deployments. Arbitrum Orbit allows developers to deploy custom L3 chains on top of Arbitrum — creating an entire ecosystem of application-specific chains. The ARB governance token was distributed via one of the largest airdrops in crypto history (March 2023), giving the community control over the protocol's future.

TVL: $10B+ | Daily Txs: 1M+ | Token: ARB | Type: Optimistic Rollup

Optimism (OP) / Superchain

Optimistic Rollup

Optimism is the second largest Ethereum L2 by TVL and the intellectual pioneer of the optimistic rollup model. Developed by Optimism PBC (now OP Labs), it introduced the OP Stack — an open-source development framework that allows anyone to build an Ethereum-compatible rollup in minutes. The OP Stack's most consequential deployment is Coinbase's Base — which runs on the OP Stack and has become one of the fastest-growing L2s in history. This network of OP Stack chains forms the 'Superchain' — a vision of hundreds of interoperable L2s all sharing security, communication protocols, and governance. The Optimism Collective governs the protocol through a bicameral system: the Token House (OP holders) votes on protocol upgrades, while the Citizens' House distributes retroactive public goods funding — one of the most innovative governance experiments in crypto.

TVL: $6B+ | Daily Txs: 500K+ | Token: OP | Key Innovation: OP Stack / Superchain

8.3 — ZK Rollups: zkSync & StarkNet

zkSync Era (ZK)

ZK Rollup (zkEVM)

zkSync Era, developed by Matter Labs, is the leading ZK rollup with full EVM compatibility — allowing Ethereum developers to deploy existing Solidity contracts with minimal changes. It uses custom ZK-SNARKs (specifically PLONK-based proofs) to cryptographically verify every batch of transactions before posting to Ethereum. zkSync's key advantage over optimistic rollups is immediate cryptographic finality — there is no 7-day challenge period. Once a ZK proof is verified on Ethereum, the transactions are final with mathematical certainty. This enables fast withdrawals back to Ethereum mainnet. Matter Labs has an ambitious roadmap including ZK Stack (similar to OP Stack) for building application-specific ZK chains, and Hyperchains — a network of interconnected ZK chains that can communicate natively.

TVL: $400M+ | Token: ZK | Type: ZK Rollup (zkEVM) | Proof System: PLONK / SNARKs

StarkNet (STRK)

ZK Rollup (ZK-STARKs)

StarkNet, developed by StarkWare, uses ZK-STARKs (Scalable Transparent ARguments of Knowledge) — a different and in some ways more powerful proof system than the SNARKs used by zkSync. STARKs require no trusted setup (making them more decentralised) and are post-quantum secure. StarkNet uses its own programming language, Cairo — a language designed specifically for writing ZK-provable programs. While this creates a learning curve for Ethereum developers, Cairo enables more efficient proof generation and programs that would be impossible to prove efficiently in Solidity. StarkWare also developed StarkEx — a separate ZK scaling service used by dYdX (crypto derivatives), Immutable X (NFT gaming), and Sorare (fantasy sports) — demonstrating ZK technology's viability at production scale before StarkNet's public launch.

TVL: $800M+ | Token: STRK | Proof System: STARKs | Language: Cairo

8.4 — Base: Coinbase's Layer 2

Base

Optimistic Rollup (OP Stack)

Base is a Layer 2 blockchain built by Coinbase on the OP Stack, launched in August 2023. It has become one of the fastest-growing L2s in Ethereum history, driven by Coinbase's 100M+ user base, its developer-friendly infrastructure, and the explosive growth of consumer applications built on Base. Base's most notable characteristic is its focus on consumer crypto applications — social protocols (Farcaster frames), creator tools, microtransactions, and mainstream onboarding. The Friend.tech social application launched on Base and briefly generated more daily transactions than all of Ethereum mainnet. Coinbase has committed to donating a portion of Base's revenue to the Optimism Collective and Ethereum public goods — aligning its economic incentives with the broader Ethereum ecosystem. Base does not have its own token, using ETH for gas — keeping economic alignment with Ethereum.

TVL: $1.5B+ | Daily Txs: 2M+ | Token: None (ETH for gas) | Operator: Coinbase

8.5 — Polygon: The Multi-Chain Network

Polygon (POL / MATIC)

ZK Rollup + Sidechain Ecosystem

Polygon is one of Ethereum's oldest and most used scaling solutions — though its architecture has evolved dramatically. Originally a PoS sidechain (Polygon PoS) with its own validator set, Polygon has pivoted to become a ZK-focused ecosystem with Polygon zkEVM as its flagship product. Polygon's historical significance is enormous: it was the first scaling solution to achieve true mass adoption, hosting Reddit's blockchain community points, Nike's NFT platform (.SWOOSH), Starbucks Odyssey (loyalty NFTs), and hundreds of other enterprise and consumer applications. At its peak, Polygon PoS processed more daily transactions than Ethereum mainnet. Polygon's AggLayer — a unified ZK-based interoperability layer — aims to connect multiple ZK chains into a single unified liquidity layer, enabling seamless cross-chain transactions without bridges. The MATIC token was rebranded to POL in 2024 as part of the ecosystem's evolution.

TVL: $1B+ | Daily Txs (PoS): 3M+ | Token: POL (formerly MATIC) | Enterprise Users: 100+

8.6 — L2 Ecosystem Comparison Table

L2	TYPE	TVL	AVG TX FEE	TOKEN	KEY STRENGTH
Arbitrum	Optimistic	$10B+	<$0.01	ARB	Largest ecosystem & DeFi TVL
Optimism	Optimistic	$6B+	<$0.01	OP	OP Stack / Superchain vision

Base	Optimistic	$1.5B+	<$0.01	None	Coinbase integration & consumer apps
Polygon zkEVM	ZK	$500M+	<$0.01	POL	Enterprise adoption & AggLayer
zkSync Era	ZK	$400M+	<$0.01	ZK	Full EVM compatibility + ZK speed
StarkNet	ZK	$800M+	<$0.01	STRK	ZK-STARKs, Cairo language
Linea	ZK	$500M+	<$0.01	None	ConsenSys / MetaMask integration
Scroll	ZK	$300M+	<$0.01	SCR	EVM-equivalent ZK rollup

09 — COMPETITIVE LANDSCAPE

Ethereum faces competition from every direction — faster chains below, modular chains alongside, and Bitcoin above as the store of value king. Understanding how Ethereum compares to each major competitor reveals where its true moats lie and where it remains genuinely vulnerable.

9.1 — Ethereum vs Solana

The Ethereum vs Solana debate is the most important rivalry in blockchain today — representing two fundamentally different philosophies about how to build a high-performance decentralised computing platform.

Where Solana Wins

- **Base layer speed:** Solana's 65,000+ TPS vs Ethereum's 15-30 TPS makes Solana dramatically faster for applications that need real-time throughput at the base layer.
- **Transaction cost:** Solana's sub-cent fees vs Ethereum mainnet's $0.50-$50+ fees make Solana more accessible for everyday users and micropayment use cases.
- **User experience:** 400ms confirmation vs 12 seconds creates a meaningfully better UX for consumer applications, games, and payments.
- **Single-shard composability:** Everything on Solana is on one chain — atomic composability across all protocols without bridges or L2 fragmentation.

Where Ethereum Wins

- **Decentralisation:** 800,000+ validators vs Solana's ~2,000. Ethereum's validator set is orders of magnitude more decentralised.
- **Security track record:** Ethereum has never experienced a network halt in 10 years. Solana has had multiple outages.
- **Developer ecosystem:** Solidity has more developers, tools, tutorials, audit firms, and institutional support than Rust-based Solana development.
- **DeFi depth:** Ethereum's $50B+ TVL vs Solana's $5B+ TVL. The deepest liquidity, most protocols, and most institutional capital are on Ethereum.
- **Institutional trust:** BlackRock, JPMorgan, Visa, and major financial institutions have chosen Ethereum for their blockchain deployments — not Solana.
- **L2 ecosystem:** Ethereum's Layer 2 networks collectively process more transactions per second than Solana — with Ethereum-level security.

The Philosophical Divide

Ultimately, Ethereum and Solana represent different bets: Ethereum bets that modular scaling (L2s) will deliver scalability without sacrificing decentralisation. Solana bets that hardware will improve fast enough that a monolithic high-performance chain can achieve all three properties simultaneously. Both approaches have merit — and the market may support both long-term.

9.2 — Ethereum vs BNB Chain

BNB Chain (formerly Binance Smart Chain) is Ethereum's most used competitor by raw transaction count — processing 3-5 million daily transactions versus Ethereum mainnet's 1-1.5 million. However, the comparison is more nuanced than raw numbers suggest.

- **Centralisation:** BNB Chain has only 21 active validators — all controlled by Binance or Binance-approved parties. It is functionally a centralised database with blockchain aesthetics, not a decentralised network.
- **EVM compatibility:** BNB Chain is fully EVM-compatible — making it easy for Ethereum projects to deploy on BNB Chain. Most major Ethereum DeFi protocols have BNB Chain versions.
- **Lower fees:** BNB Chain's fees are significantly lower than Ethereum mainnet — typically $0.10-$0.50 per transaction — making it accessible to users priced out of Ethereum.
- **Reputation:** BNB Chain has been associated with a disproportionate share of DeFi hacks, rug pulls, and scam projects — damaging its reputation among serious developers and institutional users.
- **Regulatory risk:** As Binance faces ongoing regulatory challenges globally, BNB Chain's centralised nature means regulatory action against Binance could directly impact the chain.

9.3 — Ethereum vs Avalanche

Avalanche is Ethereum's most architecturally sophisticated competitor — offering EVM compatibility, fast finality (~1-2 seconds), and the unique Subnet architecture that allows application-specific blockchains. Avalanche has positioned itself as the enterprise and institutional blockchain of choice, landing partnerships with Deloitte, AWS, and various government entities.

- **EVM compatibility:** Avalanche's C-Chain is fully EVM-compatible, allowing Ethereum projects to deploy easily. This has brought Aave, Curve, Uniswap V3, and other major protocols to Avalanche.
- **Subnet advantage:** Avalanche's Subnet architecture is genuinely differentiated — allowing enterprises to run private or semi-private blockchains with customisable rules while sharing validator security with the main network.
- **TVL gap:** Avalanche's $700M+ TVL remains far behind Ethereum's $50B+ — indicating that despite technical advantages, Ethereum's network effects are very difficult to overcome.
- **AVAX token utility:** AVAX is required to pay fees on all Subnets and to stake as a validator — creating genuine token demand. But the tokenomics are less compelling than ETH's fee-burn model.

9.4 — Ethereum vs Sui & Aptos

Sui and **Aptos**, built by former Meta engineers using the Move programming language, represent the most credible next-generation technical challengers to Ethereum. Their object-centric execution models and DAG-based consensus architectures offer genuine technical innovations — particularly in parallel transaction processing and formal verification of smart contract safety.

However, both chains face the same challenge as every Ethereum competitor: Ethereum's network effects are extraordinarily powerful. The depth of DeFi liquidity, the breadth of the developer ecosystem, the institutional relationships, and the regulatory clarity that the ETH ETF approval provided — these advantages compound over time and are extremely difficult for newer chains to replicate, regardless of technical superiority.

9.5 — Overall Comparison Table

METRIC	ETHEREUM	SOLANA	BNB CHAIN	AVALANCHE
Base TPS	~15-30	65,000+	~2,000	~4,500
With L2s	100,000+	N/A	Limited	~20,000+
Finality	~12.8 min	~400ms	~3s	~1-2s
Validators	800,000+	~2,000	21	~1,200
DeFi TVL	$50B+	$5B+	$5B+	$700M+
Avg Tx Fee	$0.50-$50	<$0.001	$0.10-0.50	<$0.10
L2 Fees	<$0.01	N/A	N/A	<$0.01
Smart Contract	Solidity	Rust	Solidity	Solidity
EVM Compatible	Native	No (SVM)	Yes	Yes (C-Chain)
ETF Approved	Yes	Pending	No	No
Founded	2015	2020	2020	2020

10 — FUTURE ROADMAP & OUTLOOK

Ethereum's development roadmap — articulated by Vitalik Buterin as five phases — represents the most ambitious and carefully planned technical upgrade roadmap in blockchain history. Each phase addresses a specific set of limitations, collectively targeting a future Ethereum that is faster, cheaper, more private, and more decentralised than anything that exists today.

PHASE	NAME	KEY UPGRADES	STATUS
The Merge	Consensus	PoS transition	COMPLETE (Sept 2022)
The Surge	Scalability	Danksharding, full data availability	IN PROGRESS
The Scourge	MEV & Censor	PBS, inclusion lists, FOCIL	RESEARCH PHASE
The Verge	Verification	Verkle trees, stateless clients	IN DEVELOPMENT
The Purge	Simplicity	History expiry, state expiry	FUTURE
The Splurge	Everything else	Account abstraction, EVM improvements	ONGOING

10.1 — The Surge: Danksharding

The most impactful upcoming upgrade for Ethereum users is **Full Danksharding** — the completion of the data availability scaling roadmap that began with EIP-4844 (Proto-Danksharding) in March 2024.

- **Proto-Danksharding (EIP-4844, DONE):** Introduced blob transactions — a new data type that L2s use to post transaction data cheaply. Reduced L2 fees by 90-99%.
- **Full Danksharding (UPCOMING):** Scales blob capacity from 3-6 blobs per block (current) to 64+ blobs per block using Data Availability Sampling (DAS). This will reduce L2 costs by another 10-100x and enable L2s to scale to millions of TPS while settling on Ethereum.
- **Data Availability Sampling (DAS):** Allows nodes to verify that blob data is available without downloading it all — by randomly sampling small pieces and using erasure coding to reconstruct the full data if needed. This enables light nodes to contribute to data availability verification.
- **Impact:** Full Danksharding is expected to enable the combined Ethereum + L2 ecosystem to process **100,000+ TPS** with sub-cent fees while maintaining Ethereum's decentralisation — the ultimate resolution of the scalability trilemma.

10.2 — The Verge: Verkle Trees

The Verge addresses one of Ethereum's deepest technical limitations: the need for validators to store and maintain the complete Ethereum state (all account balances and contract storage). As Ethereum grows, the state database grows — currently requiring ~1TB+ for a full archive node.

- **Verkle Trees:** A replacement for Ethereum's current Merkle Patricia Trie data structure. Verkle Trees use vector commitments (specifically, polynomial commitments) to create much smaller proofs of state membership — enabling stateless clients.
- **Stateless Clients:** With Verkle Trees, validators will not need to store the entire Ethereum state to validate new blocks. Instead, each block will include 'witnesses' — cryptographic proofs of all the state accessed by that block's transactions. Validators verify the witnesses without needing local state.
- **Impact:** Stateless clients will allow anyone to run a validator on a mobile phone or Raspberry Pi — dramatically lowering the hardware barrier and potentially multiplying the validator count by orders of magnitude. This would make Ethereum the most decentralised PoS network in existence.

10.3 — The Purge & The Splurge

- **The Purge — History Expiry:** Ethereum currently requires nodes to store the complete blockchain history going back to genesis (2015). EIP-4444 proposes that nodes only need to store the last year of history, with older data available from a distributed archival network. This dramatically reduces the storage requirements for running a node.
- **The Purge — State Expiry:** Ethereum state (account balances, contract storage) grows continuously. State expiry mechanisms would archive state that hasn't been accessed for an extended period, reducing the active state size validators must maintain.
- **The Splurge — Account Abstraction (ERC-4337):** Already live on Ethereum and L2s, account abstraction allows smart contracts to function as user accounts — enabling gasless transactions (sponsored by dApps), social recovery (recover wallet access via friends), batch transactions, and custom authentication methods. This is the most significant UX improvement in Ethereum's history.
- **The Splurge — EVM Improvements:** The EVM Object Format (EOF) restructures EVM bytecode for better performance and safety. Future EVM upgrades will improve computational efficiency and enable new cryptographic operations (especially ZK-friendly precompiles).

10.4 — Account Abstraction (ERC-4337)

Account Abstraction is arguably the most user-impactful upgrade in Ethereum's recent history. Deployed via ERC-4337 without requiring a protocol change, it allows smart contract wallets to replace traditional externally owned accounts — unlocking a dramatically better user experience:

- **Gasless transactions:** dApps can pay gas on behalf of users — users never need to hold ETH to use Ethereum applications.

- **Social recovery:** Lost your private key? With account abstraction, you can designate trusted friends or devices as guardians who can help recover wallet access — no more 'lost seed phrase = lost funds.'
- **Batch transactions:** Approve and swap in a single transaction instead of two. Any sequence of actions can be batched atomically.
- **Session keys:** Grant a game or app temporary, limited permissions to sign transactions on your behalf — without exposing your main private key.
- **Custom authentication:** Log into Ethereum using Face ID, fingerprint, or any other authentication method — not just a private key.

10.5 — Institutional & RWA Growth Outlook

The tokenisation of real-world assets (RWA) on Ethereum represents what many analysts believe to be the most significant long-term growth driver for the Ethereum ecosystem. BlackRock's Larry Fink has called tokenisation 'the next generation for markets' — and Ethereum is the platform of choice for virtually every major institution exploring RWA tokenisation.

ASSET CLASS	CURRENT ON-CHAIN VALUE	10-YEAR POTENTIAL
US Treasury Bills	$3B+	$500B-$5T
Money Market Funds	$1B+	$2-10T
Private Credit	$500M+	$1-5T
Real Estate	$200M+	$5-20T
Equities/Stocks	$100M+	$10-50T
Commodities	$500M+	$1-5T
Total RWA (estimate)	$5B+	$16-95T

If even a fraction of the world's $500 trillion+ in traditional financial assets moves on-chain — and Ethereum captures the majority of that market — the implications for ETH demand (as the gas token for all these transactions) are extraordinary. This is why many institutional analysts consider ETH one of the most asymmetric investment opportunities in traditional asset markets.

11 — CONCLUSION

Ethereum is the most consequential technology project of the 21st century's first quarter. In a decade, it has evolved from a teenage programmer's whitepaper into the foundational infrastructure of a new global financial system — one that operates without borders, without permission, and without central control.

What Ethereum Has Proven

• **Smart contracts work:** Despite scepticism, Ethereum has demonstrated that self-executing, trustless financial contracts can operate at scale — processing trillions of dollars in transactions without a single custodian.

• **Decentralisation scales:** With 800,000+ validators across the globe, Ethereum has proven that a maximally decentralised system can survive market crashes, regulatory pressure, hacks, and technical failures.

• **Governance works (imperfectly):** The DAO fork, The Merge, EIP-1559, and dozens of other contentious upgrades demonstrate that Ethereum's rough consensus governance — slow, messy, and heated — ultimately delivers.

• **Institutional adoption is real:** BlackRock, JPMorgan, Visa, PayPal, and dozens of other global institutions have not just invested in ETH — they have built products on Ethereum. This is the strongest possible validation of the technology.

What Ethereum Still Must Prove

• **Scalability delivery:** Full Danksharding and the complete L2 scaling roadmap must deliver on their promises. The vision of 100,000+ TPS with sub-cent fees at Ethereum-level security is compelling — but undelivered promises erode trust.

• **L2 fragmentation resolution:** The current ecosystem of dozens of L2s with fragmented liquidity and complex bridging is a genuine UX problem. The Superchain, AggLayer, and other interoperability solutions must mature.

• **Staking decentralisation:** The Lido concentration issue must be resolved — either through market forces, protocol changes, or social coordination — to maintain Ethereum's decentralisation credibility.

• **Competing with Solana on UX:** For consumer applications, Solana's speed and cost profile remains superior at the base layer. Ethereum must prove that L2s can deliver comparable UX without the complexity.

Final Assessment

Ethereum occupies a unique and largely unassailable position in the blockchain landscape. It is simultaneously the most decentralised, most battle-tested, most institutionally adopted, and most developer-rich programmable blockchain in existence. Its network effects — accumulated over a decade of being the first mover in smart contract platforms — are deep, multi-dimensional, and extraordinarily difficult to replicate.

The greatest threat to Ethereum is not technical failure but execution risk: the risk that the scaling roadmap takes too long, that L2 fragmentation permanently damages user experience, or that a competitor delivers a better experience fast enough to capture the next generation of blockchain users before Ethereum's upgrades are complete.

Final Verdict

Ethereum is the internet's settlement layer — the blockchain that every other chain either tries to replace or ultimately connects back to.

Its decade of uninterrupted operation, its institutional relationships, its developer community, and its scaling roadmap position it as the most likely foundation for the on-chain financial system of the next decade.

For researchers, developers, and investors, Ethereum remains the single most important blockchain to understand deeply — not because it is perfect, but because it is where the future of finance is being built.

Prepared by Nasir Bello | June 2025

12 — REFERENCES & SOURCES

Primary Sources

- Buterin, V. (2013). Ethereum Whitepaper. ethereum.org/whitepaper
- Wood, G. (2014). Ethereum Yellow Paper. gavwood.com/paper.pdf
- Ethereum Foundation. (2015-2025). Ethereum Documentation. ethereum.org/docs
- Ethereum Foundation. (2021-2025). EIPs Repository. eips.ethereum.org
- Ethereum Foundation. (2022). The Merge Documentation. ethereum.org/merge
- Buterin, V. (2022-2025). Ethereum Roadmap Blog. vitalik.ca

Research & Data

- DefiLlama. (2022-2025). Ethereum DeFi TVL Data. defillama.com
- Dune Analytics. (2022-2025). Ethereum On-chain Analytics. dune.com
- Messari. (2022-2025). Ethereum Protocol Research. messari.io
- Electric Capital. (2023-2025). Developer Report. developerreport.com
- Ultrasound Money. (2022-2025). ETH Supply & Burn Data. ultrasound.money
- Rated Network. (2022-2025). Ethereum Validator Data. rated.network
- L2Beat. (2022-2025). Layer 2 TVL & Security Analysis. l2beat.com
- Galaxy Digital Research. (2023-2025). Ethereum Staking Reports. galaxy.com/research

News & Institutional Reports

- BlackRock. (2024). BUIDL Fund Announcement. blackrock.com
- SEC. (2024). Spot Ethereum ETF Approval Orders. sec.gov
- Visa. (2021-2023). Crypto & Blockchain Reports. visa.com/crypto
- JPMorgan. (2022-2025). Onyx Blockchain Reports. jpmorgan.com/onyx
- CoinDesk. (2015-2025). Ethereum Coverage. coindesk.com
- The Block. (2020-2025). Ethereum Research & News. theblock.co
- Bankless. (2020-2025). Ethereum Deep Dives. bankless.com
- Unchained Podcast. (2020-2025). Ethereum Founder Interviews. unchainedpodcast.com

Technical Papers

- Buterin, V. et al. (2020). Combining GHOST and Casper. arxiv.org/abs/2003.03052

- Ethereum Research. (2020-2025). EthResearch Forum. ethresear.ch
- Flashbots. (2021-2025). MEV-Boost & PBS Research. flashbots.net
- EIP-1559: Fee market change. eips.ethereum.org/EIPS/eip-1559
- EIP-4844: Proto-Danksharding. eips.ethereum.org/EIPS/eip-4844
- EIP-4337: Account Abstraction. eips.ethereum.org/EIPS/eip-4337

This concludes the Ethereum Deep-Dive Research Series — Files 1, 2, and 3.

Prepared by **Nasir Bello** · June 2025 · For Informational Purposes Only